

Word-Sentence-Doc Embedding

1 Limitation of BOW and TF-IDF

- Vector size : big.
- Consume to much memory.
- Sparse representation (Most values are 0)
- Non similar vector representations for similar words : don't capture the meaning of words

Word embedding capture the meaning of words : similar words have similar representations.

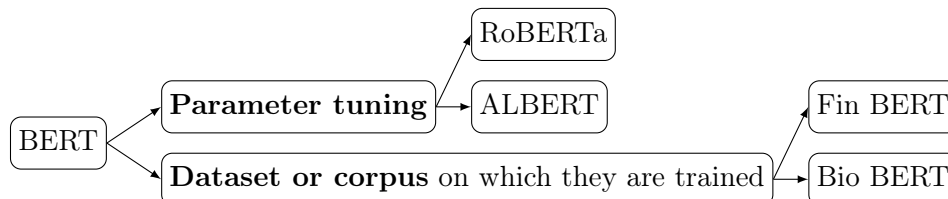
Dimension is low (200 or 300 are popular vector sizes).

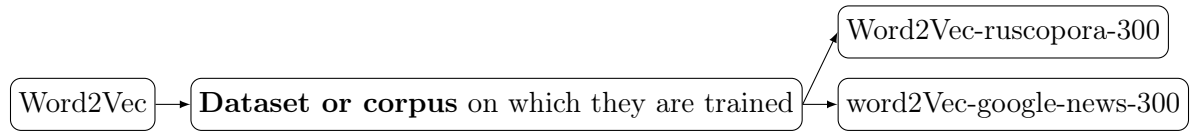
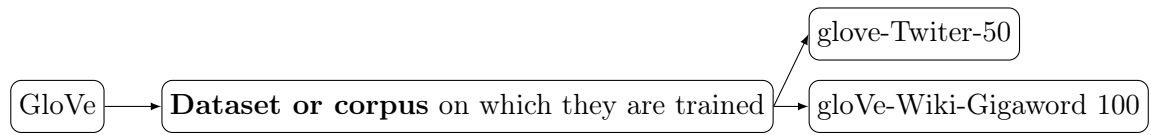
2 Word embedding techniques

CBOW, Skip-gram	Based on Transformer Architecture	Based on LSTM
Word2Vec	BERT	ELMo
	GPT	
GloVe		
fastText		

Not only these techniques convert words to vectors, but you can do arithmetic with the words. Example : King-man+woman : Queen. By the way of features.

There are variations of these techniques. The variations depend on the datasets there are training with.





3 Coding example with spaCy

See the Tutorial